

Mid-Research Report

Does Outcome-Aware Evolution Learn?
A Controlled Comparison of a Learning-Aware Alpha Miner
Against Its Paper Baseline

September 1, 2026

1 Introduction

This system uses a large language model to invent stock-picking formulas — it proposes a formula, tests it on historical market data, and repeats. The question is whether the repetition helps. If each round’s proposals are no better than the last, the system is just a random generator with a language model attached, and the only thing improving is the best of a growing pile of lucky guesses.

We compare two versions that differ in what happens *after* a formula is tested. The baseline (ORIGINAL, the original published method) proposes, tests, and keeps everything. The treatment (MAIN) adds four things: a gate that only keeps a formula if it improves the overall portfolio; a diagnosis step that works out *why* a formula failed and tells the generator; a mechanism to inject fresh ideas when progress stalls; and a memory that carries failures across rounds. Both use the same language model, the same market (China’s CSI300), the same schedule, and ten ideas per round.

Three differences between the two runs shape how the results must be read. Each was measured, not assumed.

The same 150 formulas cost different effort (§2.2).

Both run the paper’s ten ideas per round, but ORIGINAL turns each idea into three formulas — as its published method specifies — while MAIN turns each into one. Reaching 150 formulas therefore took MAIN 161 ideas and ORIGINAL only 60. A “round” means different things in each, so progress is measured per formula, and holding the formula count equal is generous to the baseline.

The two runs were tuned on different eras (§5).

MAIN’s gate was calibrated on 2013–2015 and ORIGINAL on 2021, so a shared 2022–2025 test sits seven years after one and one year after the other. This was an oversight in the setup, and it works against MAIN, whose formulas must survive six more years of drift.

More formulas automatically score better (§4.3).

The model that combines formulas into one signal improves simply from having more inputs, regardless of their quality. A 56-formula library and a 150-formula library therefore cannot be compared directly.

The report asks four questions: does the treatment build a better portfolio (§4)? does it learn (§5)? do its breeding operators actually improve on what they breed from (§5)? and does it deliver the variety and trading capacity it was designed for (§6–7)?

What we found. At equal formula count MAIN’s library builds a better portfolio than ORIGINAL’s, and over 2017–2025 it leads on both backtests (+35.5% against +20.9% a year under simple costs; +9.6% against +3.1% under realistic ones) with a higher prediction score in all nine years and a smaller worst-case loss. It also has by far the best trading capacity: run as a fund that starts at 100M and compounds, it beats the market by 7.8% a year where the baseline manages 2.4%, and it is the last library to turn unprofitable as the fund grows.

Most directly, MAIN *does* get better as it goes. Measured formula by formula — the one axis both systems share, since each mined exactly 150 — the prediction quality of what MAIN proposes rises by 0.0236 Rank IC per hundred formulas, against +0.0038 for the baseline: about six times faster. MAIN starts level with ORIGINAL (Rank IC 0.031 against 0.032 over the first fifty) and finishes 53% ahead (0.055 against 0.036). Its breeding also beats the formulas it breeds from 53–61% of the time, where the baseline’s does so *less* often than chance.

The system’s weak point is not the formulas. Over 2017–2025 they predict well and consistently under a simple portfolio construction, yet the realistic-cost path turns negative after 2020 — and the two paths disagree on the *prediction* score too, which no cost model can explain. The component that combines many formulas into one signal is where the value is lost, and that is what to fix next.

2 Experimental Design

2.1 The three libraries

Both systems use the same language model, the same market, and the same schedule. ORIGINAL keeps every formula it produces. MAIN keeps a formula only if it improves the portfolio, so it ends up with two collections: everything it generated, and the subset it chose to keep. Table 1 names all three.

Name	From	Count	What it is
MAIN-full	MAIN	150	every formula MAIN generated
MAIN-kept	MAIN	56	the ones its quality gate kept — what it would trade
ORIGINAL	ORIGINAL	150	every formula ORIGINAL generated (it keeps all)

Table 1: The three collections of formulas compared throughout. **Conclusion: the three isolate different questions** — ORIGINAL against MAIN-full tests the generators head to head at equal size, and MAIN-full against MAIN-kept tests whether the quality gate earns its keep.

Three comparisons follow. ORIGINAL against MAIN-full asks whose *generator* is better (same count, neither filtered). MAIN-full against MAIN-kept asks whether the *gate* adds value. And MAIN-kept against ORIGINAL asks how the portfolio MAIN would actually trade compares with the baseline’s whole pile.

2.2 The same 150 formulas cost very different effort

Both systems are set up identically and both generate ten ideas per round — the budget the published method specifies. The difference, quantified in Table 2, is what an idea produces: ORIGINAL turns each into three formulas, MAIN into one.

This is deliberate on both sides. ORIGINAL is run *exactly as its paper specifies*, three formulas per idea included, because the point of a baseline is to represent the published method rather than a modernised version of it. MAIN spends the same budget on one formula per idea, so that a failed

idea is a failed *idea* rather than one of three variations that might succeed by chance. Neither is a defect in the setup; the consequences below simply follow from it.

	MAIN	ORIGINAL
Ideas per round	10	10
Formulas per idea	0.93	2.50
Rounds to reach 150	18	6
Distinct ideas explored	161	60
Deepest breeding chain	5	5

Table 2: Both systems run the paper’s budget of ten ideas per round; they differ in how many formulas an idea yields, with ORIGINAL following its published three-per-idea specification. **Conclusion: reaching the same 150 formulas therefore cost main $2.7\times$ as many distinct ideas — it spends its budget on new thinking rather than on variations of the same thought.**

Two consequences follow. A “round” is not a comparable unit (ten formulas in MAIN, thirty in ORIGINAL), so progress is measured per formula and each system is only ever compared with itself. And fixing the count at 150 is generous to ORIGINAL, which reaches it having committed to far fewer distinct ideas — while the merging model separately rewards simply having more formulas (§4.3).

2.3 Backtests and the no-leak window

Every library is tested twice, and both tests buy the same portfolio — the 50 highest-scoring stocks each day, equally weighted — so the only differences are how the formulas are merged and how trading is charged.

1. **The simple backtest** merges the formulas with a single model and charges a flat trading fee, regardless of how much money is being managed. This is the setup the original paper used.
2. **The realistic backtest** merges them with a two-stage model and charges a fee that *grows with position size*, reflecting the fact that large trades move prices against you. This is the one to believe for questions about real profitability.

Remark 1 (Neither system saw the test period). *Everything is scored on years that came after any data either system used to choose its formulas. ORIGINAL selected on 2021 and earlier and in any case never filtered anything; MAIN’s gate used data ending in 2015. Both backtests then train their own merging model on earlier years before scoring the test period, so no result below is contaminated by hindsight.*

3 What the Numbers Mean

A *formula* scores every stock each day; the *portfolio* is built by buying the 50 highest-scoring ones. Two questions matter: does the score predict what happens next, and does trading on it make money after costs? Table 3 defines every measure used in this report, with the range it can take, what counts as good, and what MAIN achieves.

Two ways of quoting a return. *Excess* return subtracts the market, answering “did the stock-picking add anything?”. *Net* return adds the market back, answering “what did the money actually earn?”. Every return table below shows the market’s own return too, so either can be read off. The two backtests measure the market slightly differently — the realistic-cost one corrects for dividends, the simple one does not — so their return columns are compared *between libraries*, not against each other.

4 Results

All numbers are computed on **2022-01-01 – 2025-12-26**, out of sample for both arms (§2.3). Two independent backtests are reported because they disagree, and the disagreement is informative.

4.1 Per-year results

Every library is scored by both backtests on 2022–2025, reported year by year: Table 4 under simple costs, Table 5 under realistic ones.

The two backtests disagree about which library is best, and the disagreement is the point. Under simple costs MAIN-full leads in every year. Under realistic costs ORIGINAL edges ahead on prediction in three of four years while MAIN-full still leads on return — MAIN’s formulas are cheaper to trade, so they keep more of a smaller edge. Both agree that 2025 was the worst year for every library, and that the 56-formula gated set scores below both 150-formula libraries, for the reason §4.3 explains.

One correction was needed first. The realistic-cost setup measured the market using an index that *excludes* dividends, while the portfolio’s own returns *include* them — which silently credited the strategy with the market’s dividend yield (about 4.25 percentage points a year) as if it were skill. Correcting it moved the gated set’s return from -2.4% to -5.3% and the null model from -11.0% to -13.5% . The prediction scores did not move at all, since they compare stocks against each other on the same day and so never see a market-wide effect.

What the null model is for. The *null model* row runs the identical machinery — the same merging model, the same top-50 portfolio, the same cost charges — on the base price and volume data alone, with *no mined formula in it*. It exists to answer an obvious objection: if the whole pipeline made money, how would we know the edge came from the discovered formulas rather than from the machinery that trades them? The null model measures exactly that machinery on its own. It returns -13.45% a year, so the machinery by itself destroys value; anything above that line is attributable to the formulas.

The headline. No library makes money on 2022–2025. All three still beat the null model by 8–11 percentage points, so the formulas do carry real information — the machinery is not the source of the edge, and in fact subtracts from it. Over 2017–2025 (§4.2) the same libraries are profitable for four consecutive years, so this negative result belongs to the recent period rather than to the method.

4.2 The long window: 2017–2025

The 2022–2025 window is short. Both backtests were re-run over 2017–2025 (Table 6), with the merging model fitted on 2005–2012 and checked on 2013–2015. This window overlaps ORIGINAL’s own 2021 test year, so it favours the baseline rather than MAIN.

The two backtests disagree, and that locates the problem. Under simple costs MAIN-full makes money in all nine years and predicts better than ORIGINAL in every one. Under realistic costs the same library is profitable only through 2020 and loses money from 2021.

Costs cannot explain a gap that large, because the two also disagree on the *prediction* score — and prediction has no cost model in it at all. On identical formulas over identical days, the simple path measures 0.09–0.13 after 2020 while the realistic path drops to 0.03–0.06. Nor is it how the portfolio is built: both buy the top 50 stocks, equally weighted, with no position limits. The one thing that differs is the *combiner* — the model that merges many formulas into a single score. Since prediction is measured on the combiner’s output, that is what the gap points to.

This is not evidence that the strategy is profitable; the realistic path is the one to believe, and it says otherwise after 2020. It is evidence that the problem is not the formulas: they predict well and steadily across the whole period when merged by the simpler model. The combiner is what to fix before judging the formulas’ economic worth.

4.3 The count confound, and what it hides

Taken at face value, Table 5 says ORIGINAL predicts best (Rank IC 0.052) ahead of MAIN-full (Rank IC 0.049), with the 56-formula kept set last (Rank IC 0.038) — so being selective looks like a mistake. That comparison is invalid. The combiner gets one input per formula and improves simply from having more inputs, whether or not the extra ones are any good. Scoring random subsets of a *single* library makes this obvious (Table 7):

Holding the count fixed at 56 — the size of MAIN’s kept set — and drawing five random 56-formula subsets of each 150-formula library gives the like-for-like comparison in Table 8:

Two things follow. First, on the *generator* question — comparing the two unfiltered libraries at equal size — MAIN wins on both measures: prediction (Rank IC 0.046 against 0.040) and annual return (−4.80% against −7.43%, a 2.6-point lead), with no overlap across the five draws. ORIGINAL’s apparent lead at full size was purely the count effect.

Second, on the *gate* question, the right comparison is against what the gate is for. It selects formulas that improve the portfolio’s *return*, not formulas that predict best in isolation. Against the baseline at the same size it does exactly that: it gives up 0.002 of prediction (Rank IC 0.038 against 0.040) and gains 2.2 points of annual return (−5.28% against −7.43%). Judged on the quantity it optimises, the gate works.

Its limitation is narrower than that trade suggests. Against a random 56 drawn from MAIN’s *own* library the gate is roughly a wash — slightly behind on both prediction (Rank IC 0.038 against 0.046) and return (−5.28% against −4.80%). So the gate adds real value over the baseline’s formulas, but it does not yet beat simply taking a random handful of MAIN’s own. Given that its selections were made under a merging model whose prediction quality collapses after 2020 (§4.2), that is what one would expect: the gate is choosing well against a scoring function that is itself degraded. Fixing the merging model is the prerequisite for testing the gate properly.

One measure does favour ORIGINAL: judged one formula at a time rather than as a portfolio, its formulas are stronger (average Rank IC of 0.013 against 0.011). That is the same trade seen from the other side — MAIN’s formulas combine better, ORIGINAL’s stand alone better.

5 Does the System Learn?

5.1 Definition, and how it is measured fairly

A generator cannot produce a strictly better formula every time; demanding that sets a bar no search meets. The definition, fixed before measurement:

Definition 1 (Learning). *A system learns from its mistakes and successes if, as the search progresses, (L1) it does not get worse — the quality of what it proposes is flat or rising; (L2) the set it keeps improves; and (L3) the best result so far keeps rising.*

Two choices make the comparison fair, both following from the fact that ORIGINAL is run exactly as its paper specifies (§2.2) rather than being adjusted to match MAIN.

Progress is measured per formula, not per round. Because the published method yields three formulas per idea and MAIN yields one, a round is ten formulas in MAIN and thirty in ORIGINAL — so anything “per round” would compare different quantities. Both libraries contain exactly 150 formulas mined in order, which is a common axis.

Each system is scored on the metric it recorded itself, in its own library file, on its own evaluation window. Absolute levels are therefore never compared across systems — only the *slope* of each system against itself, which is precisely the question “does it get better as it goes?”. Both libraries record prediction quality (Rank IC) per formula, so that is the shared measure.

5.2 Do the formulas get better?

Table 9 is the central evidence that the system learns. MAIN begins level with the baseline — its first fifty formulas average a Rank IC of 0.031 against ORIGINAL’s 0.032 — and ends 53% ahead (0.055 against 0.036). The rank correlation between formula number and quality is +0.83 for MAIN against +0.44 for ORIGINAL, so the improvement is steady rather than driven by a few late outliers. Figure 1 plots it.

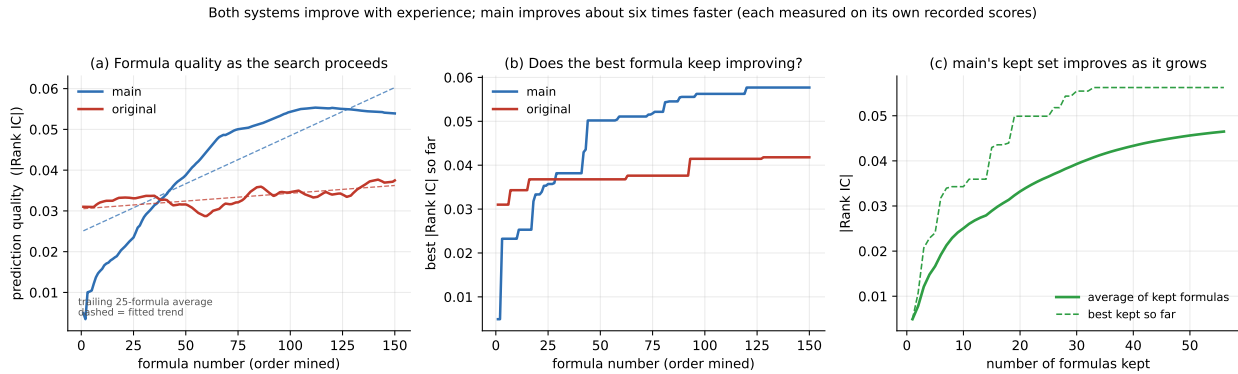


Figure 1: Left: formula quality against the order formulas were mined (trailing 25-formula average, dashed lines the fitted trends). Middle: the best formula found so far. Right: MAIN’s kept set as it grows.

Two caveats. MAIN’s size-neutralised prediction score — which strips out any part of the edge explained by simply favouring small stocks — rises more modestly and is not statistically distinguishable from flat (+0.0028 per 100 formulas, $p = 0.31$), though its first-to-last-third comparison is positive (0.0201 $\rightarrow$ 0.0248, $p = 0.041$). And formula quality here is each system’s own recorded

score, so the two columns answer “did this system improve?” and not “whose formulas are better?” — that question is answered at equal count in §4.3.

5.3 Lineage: do the operators improve on their parents?

New formulas are bred from old ones, so we can ask directly whether a child beats its parent. Both systems record this fully (MAIN 161 formulas with 155 parent links, ORIGINAL 60 with 37). A “mutation” has one parent, so beating it is a coin flip at 50%; a “crossover” has two, so beating *both* is about 33% by chance. Parents are already the stronger formulas in both systems, which makes the true bar lower still.

Table 10 is the sharpest contrast in the report. Measured on prediction quality (Rank IC), MAIN’s mutation produces a better formula than its single parent 61.3% of the time against a 50% coin-flip bar, and its crossover beats *both* parents 53.5–58.1% of the time against a 33% bar — roughly $1.7\times$ chance. ORIGINAL’s mutation manages 27.6% on the same measure, so its bred formulas are usually *worse* than what they came from. That is what breeding without reading the results looks like.

5.4 Verdict on the three tests

Table 11 scores all three. Both systems pass the first — neither degrades. The difference is rate: MAIN improves about six times faster, and only MAIN has a kept set whose quality can improve, because ORIGINAL keeps everything it produces.

Beyond prediction quality, MAIN records fourteen further measures the baseline never computes, and twelve improve significantly as the gate keeps formulas: its own portfolio-contribution score (U) from 0.55 to 0.62, the statistical strength of new additions (t -statistic) from 3.9 to 6.3, and the stability score from 0.50 to 0.98. Every best-so-far record rises, and all of this happens while the bar for statistical significance is *tightening*, because testing more ideas demands stronger evidence. All 29 replacements beat the formula they displaced on that same t -statistic, and no formula was ever kept whose real behaviour contradicted the economic story it claimed. Two measures worsen: formulas grow more similar to one another (the highest correlation against an existing formula rises from 0.00 to 0.68) and decay faster — the price of buying variety.

6 Trading Capacity

Market impact is the only cost term that scales with position size, so this applies to the full-cost path alone (Backtest v2 charges a flat fee, which is size-independent). Participation is the traded weight divided by the stock’s average daily volume as a weight, so raising capital raises impact quadratically. Each library was re-priced at 100M, 500M and 1B CNY on both windows.

Are profits reinvested? Yes — and it costs the strategy. The portfolio is always fully invested: each day every holding grows or shrinks with its own gain or loss, and the positions are rescaled to add up to the whole fund, so profits change *what* is held. That means a fund that makes money gets bigger, and a bigger fund pays more to trade, because its orders are larger relative to daily volume.

A run labelled “100M” could be charged two ways: as a fund permanently held at 100M, or as a fund that *starts* at 100M and compounds. Table 12 reports both. In the compounding version the fund’s value is carried forward day by day at its own realised return, and each day’s trading is charged against that running value — so growth raises the cost, which slows further growth.

Compounding costs 1–3.5 percentage points a year, and it costs the most where the fund grows the most: MAIN-full multiplies $3.6\times$ at 100M and gives up 2.3 points, and 3.4 points at 500M and 1B. The effect is self-limiting — a fund that grows pays more, which slows its growth — which is why the multiple falls from $3.6\times$ at 100M to $1.6\times$ at 1B.

One headline changes. Under the fixed-size convention MAIN-full still beat the market at 1B (+2.25%); once the fund is allowed to compound, it does not (−1.20%). Its practical ceiling is therefore somewhere between 500M and 1B rather than beyond 1B. The ordering between systems is unchanged at every size, and every library remains positive on a *net* basis at every size — MAIN-full earns +5.09% a year even at 1B — because the market itself returned 6.60% over this period.

Reading the curve. MAIN-full has the best capacity at every size, and by a wide margin: on the compounding basis it beats ORIGINAL by 5.4 percentage points of excess return at 100M, 4.6 at 500M and 4.2 at 1B. Its formulas are therefore cheaper to trade as well as more predictive. Every library loses ground as size rises — roughly 8 points of excess return between 100M and 1B, with trading cost climbing from about 4.9 to 9.4 basis points while the amount traded stays the same. The practical ceiling sits near 100M for ORIGINAL and MAIN-kept, which turn negative by 500M, and between 500M and 1B for MAIN-full.

On the shorter 2022–2025 window, nothing works at any size. There, excess returns run from −2.4% to −11.5% depending on library and size, and the market itself returned almost nothing — the equal-weight benchmark gained 1.49% a year and CSI300 0.95% (or −1.36% ignoring dividends), after falling 20.4% in 2022 and 10.1% in 2023. With no market gain to lift the portfolio, the net figures stay negative too.

7 Diversity and Regime Behaviour

MAIN’s quality gate was built to buy *variety* — it rejects a formula that closely resembles one already kept, or that adds no genuinely new information — and its feedback reports separately how each formula behaves when markets crash and when they rally. Both are testable design claims, and Table 13 tests them. Similarity is judged with the gate’s own threshold and its own correlation measure, so both libraries face the identical bar the gate applied.

Diversity: the design claim holds, modestly. At equal size MAIN-full carries more independent directions than ORIGINAL (effective rank 38.1 against 36.0, 26% against 24% of its factor count) and fewer redundant pairs above the gate’s threshold (4.4% against 5.1%). The clearest margin is in *operator evenness*: both libraries touch a similar number of operator families, but MAIN’s usage is spread far more evenly (18.3 effective operators against 13.6), meaning ORIGINAL leans much harder on a few favourites. That is what a search penalised for redundancy and rewarded for breadth should look like. The margins are real but not large, and MAIN does not lead on mean pairwise correlation.

The gated zoo is the interesting case: with only 56 factors it is markedly *more* internally correlated (0.375 mean, 15.5% of pairs above threshold) and carries an effective rank of 12.6. Selecting for marginal contribution does not produce a mutually orthogonal set — it produces a set whose members each added something *at the moment they were admitted*, against a book that was smaller then. This is the same breadth–purity cost visible in $\rho_{\max}$ rising from 0.00 to 0.68 over the run (§5).

Regime robustness: not demonstrated. Every library shows a much stronger edge on crash days than on average — crash $|IC|$ is four to five times the all-day figure, and roughly nine in ten factors retain their edge when markets fall. But MAIN does not lead: ORIGINAL’s crash $|IC|$ is slightly higher (0.0576 against 0.0540) and marginally more of its factors survive a crash (90% against 88%). Reporting crash and rally IC to the generator has not yet produced a measurably more crash-robust library. The instrumentation exists; the search has not exploited it.

8 Mechanisms

Both systems breed new formulas the same way: a first round proposes fresh ideas, then rounds alternate between *mutation* (tweak one formula) and *crossover* (combine two). The names match; what happens inside them does not. Table 14 summarises the differences described below.

8.1 Which formulas get bred

Choosing what to mutate. ORIGINAL takes every formula from the previous round, sorts them by an arbitrary index number, and mutates all of them in that order. No measure of how well a formula performed enters the choice.

MAIN sorts each formula by *what its test result said*. A formula that failed in a way the system can pin down is sent for repair — breed a child that fixes that specific defect. One that failed for no identifiable reason, or whose obvious repair has already been tried, is abandoned in favour of a fresh start. A formula that succeeded is used as breeding stock, and a share of the winners also get pushed further in the same direction. Repairing failures takes priority over pushing winners.

Choosing crossover pairs. ORIGINAL scores every possible pair on a mix of variety and average performance, so a pair can win on being different while being worse than the pair it beat.

MAIN ranks formulas by how much each one improved the overall portfolio, then breeds the two best. That ranking is deliberately cautious: when the measurement is too noisy to tell two formulas apart, the scores are pulled toward the average so the system declines to choose rather than breeding whichever looked better by luck.

8.2 Diagnosis: knowing *why* something failed

ORIGINAL has none. Its mutation prompt receives the parent formula and its numbers and asks for something different; if the language model fails to answer, it falls back on one of six hardcoded ideas chosen by keyword.

MAIN diagnoses failure and success. On **failure**, the system identifies which of five things went wrong — too expensive to trade, too similar to what it already has, fitted to noise, no real signal, or an edge that faded — and tells the generator using that formula’s own measured numbers. A formula that traded too fast to cover its costs is told exactly that, with its trading rate quoted; one that duplicated an existing formula is shown which one and by how much. The system names the problem and leaves the fix to the generator. It also pinpoints which part of the formula caused the fault, and notices when the same repair has been attempted repeatedly without success, at which point it stops refining and starts over.

On **success**, rather than mechanically splicing two formulas together, it identifies what each parent demonstrably did well and hands those pieces to the generator to build something new from. The mechanical splice was abandoned after children were found to inherit only about a fifth of the second parent — less than random mixing would give.

8.3 Fresh ideas, the quality gate, and feedback

ORIGINAL’s list of research directions is fixed for the whole run, so once a direction is exhausted the search keeps mining it anyway. MAIN injects fresh directions when progress stalls or on a schedule, builds them from what the run has learned so far, marks exhausted directions, and *adds* to the list rather than replacing it, so directions that are still working survive.

ORIGINAL has no quality gate. It checks only that a formula is not absurdly complicated; its one similarity check is switched off in the code. Every formula that runs is kept — which is why its “150 formulas” and its “kept set” are the same thing. MAIN keeps a formula only if it improves the overall portfolio, tested in layers: minimum standards on return and trading cost; proof that the improvement is larger than the measurement noise around it; checks that the formula is not a near-copy of something already held or of another formula proposed at the same time; a requirement that the formula’s real behaviour matches the economic story it claims (so a formula cannot be flipped to fit the data without rewriting its rationale); a rising bar for statistical evidence as more ideas are tested; and, to displace an existing formula, a margin of improvement rather than a hair’s breadth.

ORIGINAL’s feedback to the generator is the previous idea plus free-text commentary — no breakdown of what went wrong, no memory of past failures. MAIN returns the whole scorecard, with the worst dimension named first, plus which market conditions the formula failed in. Because each round starts with a blank slate, the lasting memory is a running summary that carries both the kept formulas *and the rejected ones with the reason they were rejected* into later rounds.

9 Trajectory Analysis

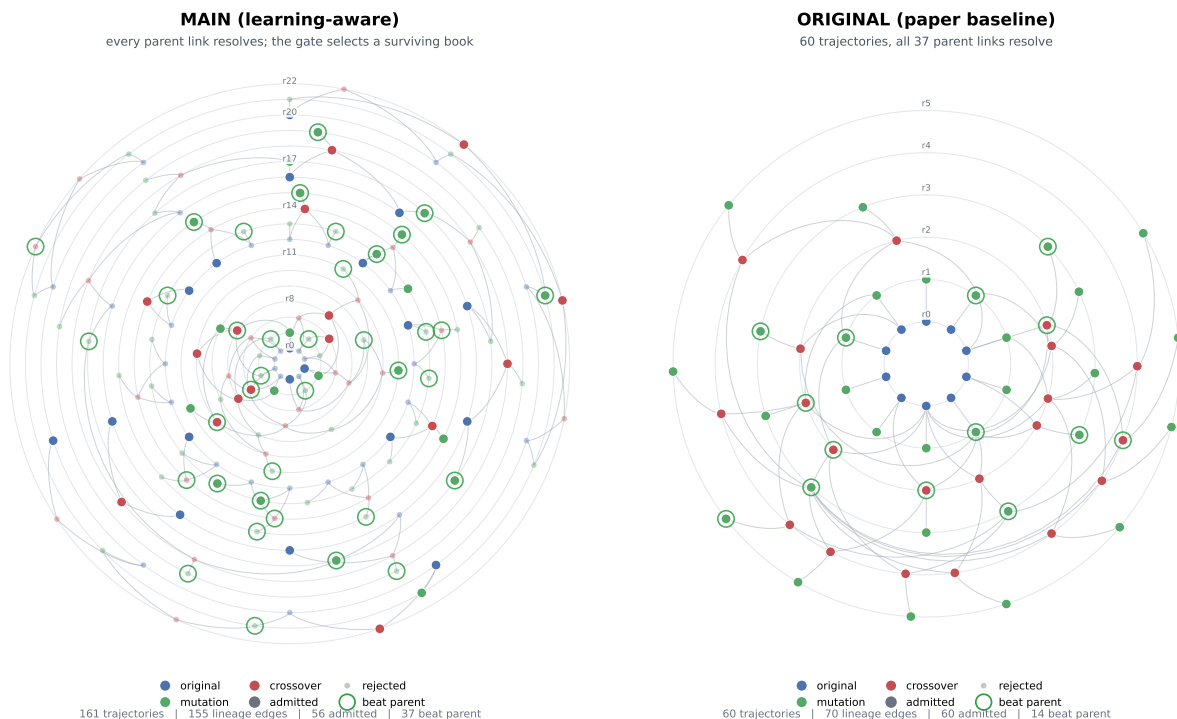
Figure 2 draws each search as a circular expanding lineage: every ring is one evolution round (radius grows outward with the round), every node is a trajectory, colour marks the operator that produced it, filled nodes were admitted to the book, a green halo marks a child that beat its best parent, and each curve is a parent→child link.

Three things are visible. MAIN’s admitted nodes are a minority scattered through every ring rather than clustered early — the gate keeps finding factors worth keeping late in the run, which is what lets the accumulated book improve (§5) while per-factor quality stays flat. MAIN runs eighteen rounds to ORIGINAL’s six, but that follows from factor yield rather than search depth (§2.2): both reach the same maximum breeding depth of five generations, so MAIN’s extra rounds buy more distinct hypotheses (161 against 60), not a longer chain of inheritance. And green halos — children beating their best parent — appear throughout MAIN’s tree on both operators (37 of 111 bred children) but are sparser in ORIGINAL (14 of 48), below what chance alone would give.

A ring does not mean the same thing in the two panels — about ten factors in MAIN, thirty in ORIGINAL — so the panels compare on structure, not ring count. MAIN’s rounds 3–7 are absent because the production run was resumed after a mid-run model rollback; rings are indexed by ordinal position, so the gap leaves no blank ring.

10 Discussion

What holds up. MAIN gets measurably better as it searches: the prediction quality of its formulas rises six times faster than the baseline’s (Table 9), starting level and finishing 53% ahead. Compared at the same number of formulas, its generator produces the better raw material. Over 2017–2025 it wins on both backtests (+35.5% against +20.9% a year under simple costs, +9.6% against +3.1%



(a) MAIN: 161 trajectories over 18 rounds, 155 lineage links, 56 admitted, 37 children beating their best parent.

(b) ORIGINAL: 60 trajectories over 6 rounds, 70 lineage links, no admission step, 14 children beating their best parent.

Figure 2: Trajectory lineage: each ring is a round, each dot a formula, each curve a parent-to-child link. **Conclusion: both searches are fully traceable, but only main’s is *selective* — its kept formulas (filled dots) keep appearing in the outermost rings, so the search is still finding things worth keeping late in the run, whereas original keeps everything and its tree simply widens.**

under realistic ones), predicts better in all nine years, loses less at its worst, and has by far the best trading capacity — beating ORIGINAL by 4–5 percentage points a year at every fund size once the fund is allowed to compound. It holds more genuinely different ideas and spreads them across a wider range of building blocks. Its selection process demonstrably learns: the portfolio it assembles improves steadily even as the statistical bar for admission rises, every one of its 29 replacements beat the formula it displaced, and it never kept a formula whose real behaviour contradicted the story it told. Its breeding beats what it breeds from 53–61% of the time, where the baseline’s does so less often than chance.

What does not. MAIN’s kept set predicts slightly worse than a random 56 drawn from its own library, and its members are more similar to each other than in either full library — the gate buys portfolio contribution at some cost in raw prediction, and against its own library that trade is currently about a wash. Reporting how formulas behave in crashes has not yet produced a more crash-resistant library. And nothing is profitable on 2022–2025 at any fund size. Note also that MAIN’s size-neutralised prediction score, which strips out any edge coming from simply favouring small stocks, improves far less clearly than its raw score.

Where the loss sits. The formulas predict well: every library scores positively every year across the whole period, and under the simple merging model MAIN-full makes money in all nine. The failure is downstream. The two backtests disagree not only on returns but on *prediction* — 0.09–0.13 against 0.03–0.06 after 2020, on identical formulas over identical days. No cost model can cause that, because prediction has no costs in it. Both build the portfolio identically. The only remaining difference is the model that merges many formulas into one score, and that is where the value is being lost.

Three traps worth remembering. Each one, uncorrected, reverses a headline. The realistic setup measured the market with an index that excludes dividends while the portfolio’s returns include them, crediting the strategy with the market’s dividend yield as skill — prediction scores are immune, since they compare stocks within a day. The two systems’ own stored numbers were computed under different cost assumptions on different years and must never be placed side by side. And the merging model improves simply from being given more formulas, so any comparison that does not hold the count fixed penalises the more selective system.

11 Which System Is Better?

main, on the evidence collected here. It is ahead on thirteen of the fifteen measures in Table 3, and ahead on the ones that matter most:

- **It learns, and the baseline barely does.** Formula quality improves six times faster (+0.0236 against +0.0038 Rank IC per hundred formulas). MAIN starts level and finishes 53% ahead.
- **Its breeding works.** New formulas beat the ones they were bred from 53–61% of the time; the baseline’s do so *less* often than chance (21–28%), meaning its breeding usually destroys value.
- **Better portfolios, on both backtests, in every year.** +35.5% against +20.9% a year under simple costs and +9.6% against +3.1% under realistic ones over 2017–2025, with a higher prediction score in all nine years and a smaller worst-case loss (−7.9% against −10.6%).
- **It scales better.** Run as a fund that starts at 100M and compounds, it beats the market by 7.84% a year against the baseline’s 2.40%, and holds that lead at every fund size.
- **It explored more.** The same 150 formulas cost MAIN 161 distinct ideas and ORIGINAL only 60, so MAIN’s result comes from genuinely wider search rather than variations on fewer thoughts.

Where the baseline wins. ORIGINAL’s formulas are marginally cheaper to trade (5.49 against 5.81 basis points) and hold their edge slightly better in market crashes (0.058 against 0.054). Judged one formula at a time on its own recorded scores its formulas also look stronger — but that is the flip side of a selection rule that deliberately prefers formulas which add something new to the portfolio over formulas that merely look good alone.

The honest qualifications. MAIN’s advantage is established on a single run of each system, so some of the gap could be luck in what the language model happened to propose; and neither system produces a strategy that is profitable under realistic costs on the most recent four years. The first is addressed by the repeated runs proposed below, the second by fixing the merging model.

12 Next Steps

Ordered by effort, quickest first.

1. **Re-run both systems on identical splits.** The seven-versus-one-year gap between tuning periods (§1) was an oversight and works against MAIN. Removing it makes every comparison in this report cleaner, and costs only the compute to re-mine.
2. **Measure generation luck: three to five runs of each system.** Every result here rests on one run apiece. Repeating both and reporting the spread establishes whether MAIN’s margin exceeds run-to-run variation — without which no single-run difference can be called real.
3. **Test on the S&P 500, two ways.** First, mine and test on US data, to see whether the advantage is a property of the method or of this market. Second, mine on CSI300 and test on the S&P without re-mining — a strict out-of-sample test of whether the formulas capture something general rather than something local.
4. **Re-mine and re-test over a longer window, on both backtests.** This settles the open decay question: does MAIN still lead over a longer horizon, and does its edge resist decay better than the baseline’s, under both simple and realistic costs?
5. **Add continuous learning and a falsification critic.** The current system learns within a run; carrying that across runs is the natural extension. A critic that actively tries to *disprove* each proposed formula would attack the weakness this report identifies — the generator’s raw output quality, the one thing that has not improved.
6. **Fix the merging model and the portfolio constructor.** The largest and hardest item, and the one that binds everything else: prediction quality is good, yet the realistic path turns negative after 2020 and the two backtests disagree on prediction itself (§4.2). Replacing the merging model with a properly specified one and the position builder with an industry-standard mean-variance construction is what stands between a demonstrated edge and a strategy worth running. Until it is done, no amount of generator improvement will show up in the returns.

Metric	What it means	Range	Good is	main	original
IC	Do the stocks it ranks highest actually go up most? Measured within each day, so market-wide moves cancel out.	−1 to 1; real 0.01–0.10	> 0.03	0.118	0.105
Rank IC	Same, using only the <i>order</i> of scores. Since we buy the top 50, this is the number that matters.	−1 to 1	> 0.03	0.114	0.101
Rank IC, one formula	The same, for a single formula rather than the whole portfolio.	−1 to 1	> 0.02	0.046	0.034
Rank ICIR	How <i>consistent</i> the edge is day to day, rather than how big.	usually 0–1	> 0.3; > 0.5 strong	0.876	0.808
Annual re- turn, simple	Compound yearly rate after a flat trading fee.	unbounded	> 5% excess	+35.5%	+20.9%
Annual re- turn, realistic	The same, once trading impact grows with size.	unbounded	> 5% excess	+9.6%	+3.1%
Info. ratio	Return per unit of risk.	unbounded	> 1; > 2 strong	4.00	2.51
Max draw- down	Worst peak-to-trough loss along the way.	−100% to 0	better than −20%	−7.9%	−10.6%
Turnover	Share of the portfolio traded daily — the multiplier on every cost.	0 to 1	< 0.20/day	0.106	0.106
Cost	Trading friction, in hundredths of a percent traded, at 100M.	≥ 0	< 10	5.81	5.49
Capacity	Excess return running a fund that starts at 100M and compounds.	unbounded	> 0	+7.84%	+2.40%
Effective rank	How many genuinely <i>different</i> ideas the library holds. 150 near-copies count as far fewer.	1 to n	> 20% of n	38 (26%)	36 (24%)
Redundancy	Share of formula pairs too similar to each other.	0 to 1	low	4.4%	5.1%
Crash IC	Does the edge survive the worst 20% of market days?	−1 to 1	$\geq$ overall IC	0.054	0.058
Improvement	Gain in formula quality per 100 formulas mined — does it learn?	unbounded	> 0	+0.0236	+0.0038

Table 3: Every measure used in this report, and what each system achieves, over 2017–2025 for the full 150-formula libraries. Return, cost and capacity are from the **realistic cost model** (trading impact grows with fund size), except the “simple” return row; prediction scores are unaffected by the cost model. **Conclusion: main is ahead on thirteen of fifteen measures** — including every measure of prediction, return, risk, capacity and learning rate. ORIGINAL leads only on trading cost (5.49 against 5.81 basis points) and on how well the edge survives market crashes (0.058 against 0.054); turnover ties. “Good is” are conventional rules of thumb, not thresholds the system enforces.

		2022	2023	2024	2025	full
IC	MAIN-kept (56)	0.0972	0.0930	0.0919	0.0796	0.0905
	MAIN-full (150)	0.1391	0.1337	0.1364	0.1116	0.1302
	ORIGINAL (150)	0.1060	0.1247	0.1195	0.0902	0.1101
Rank IC	MAIN-kept	0.0938	0.0921	0.0917	0.0756	0.0883
	MAIN-full	0.1364	0.1308	0.1339	0.1060	0.1268
	ORIGINAL	0.1024	0.1227	0.1157	0.0856	0.1066
Return	MAIN-kept	34.81%	19.73%	13.84%	1.04%	16.76%
	MAIN-full	52.07%	37.84%	38.98%	23.42%	37.74%
	ORIGINAL	29.24%	30.34%	33.41%	3.54%	23.55%
	<i>CSI300 (price)</i>	−21.7%	−11.4%	+14.7%	+17.7%	−1.4%
	<i>CSI300 (total)</i>	−20.4%	−10.1%	+16.5%	+19.1%	+0.9%

Table 4: **Simple backtest** (flat trading fee), 2022–2025, at **100M CNY** of capital. Returns are excess of the CSI300 price index, whose own return is shown for reference. **Conclusion: main-full leads original on every metric in every year**, and both beat a market that fell over this period.

		2022	2023	2024	2025	full
IC	MAIN-kept (56)	0.0321	0.0079	0.0203	0.0369	0.0242
	MAIN-full (147)	0.0433	0.0151	0.0294	0.0394	0.0318
	ORIGINAL (149)	0.0347	0.0276	0.0341	0.0408	0.0343
Rank IC	MAIN-kept	0.0433	0.0194	0.0306	0.0569	0.0375
	MAIN-full	0.0598	0.0333	0.0429	0.0613	0.0493
	ORIGINAL	0.0526	0.0447	0.0481	0.0609	0.0515
Return	MAIN-kept	−1.68%	−7.74%	−5.74%	−5.86%	−5.28%
	MAIN-full	+3.56%	−2.83%	−3.60%	−6.38%	−2.37%
	ORIGINAL	−2.87%	−1.22%	−0.75%	−9.42%	−3.62%
	<i>null model</i>	−9.81%	−24.60%	−12.40%	−5.72%	−13.45%
	<i>benchmark (EW, total)</i>	−13.0%	−4.9%	+21.3%	+20.0%	+1.5%
	<i>CSI300 (total)</i>	−20.4%	−10.1%	+16.5%	+19.1%	+0.9%

Table 5: **Realistic-cost backtest** (trading impact grows with size), 2022–2025, at **100M CNY** of capital. Returns are excess of the benchmark shown. The *null model* is the identical machinery run with no mined formula in it, so it isolates what the merging model and portfolio rule contribute on their own. **Conclusion: nothing makes money on this window, but all three libraries beat the null model by 8–11 points — the formulas carry genuine information, and the machinery that trades them subtracts from it rather than supplying the edge.**

	2017	2018	2019	2020	2021	2022	2023	2024	2025	full
<i>Simple backtest — annual return</i>										
MAIN-full	40.8	60.8	18.9	29.4	48.5	54.2	30.4	28.6	14.7	35.5
ORIGINAL	3.8	29.9	8.2	21.1	33.7	33.7	26.2	32.0	5.4	20.9
MAIN-kept	8.4	37.2	5.8	13.3	29.8	30.4	20.3	12.9	10.1	18.2
<i>Simple backtest — prediction (Rank IC)</i>										
MAIN-full	.130	.126	.102	.112	.099	.125	.114	.123	.091	.114
ORIGINAL	.108	.109	.092	.096	.093	.103	.111	.110	.084	.101
MAIN-kept	.093	.099	.077	.076	.074	.089	.071	.078	.071	.081
<i>Realistic backtest — annual return</i>										
MAIN-full	52.8	41.0	19.8	17.8	−5.9	−2.0	−5.3	−8.1	−7.0	9.6
ORIGINAL	20.4	18.7	5.1	18.4	−2.5	−7.9	−3.7	−5.4	−9.7	3.1
MAIN-kept	31.8	35.5	7.1	8.9	−5.0	−8.2	−10.1	−10.2	−6.4	3.6
<i>null</i>	−4.9	−12.2	5.7	−4.8	−7.6	−14.4	−26.5	−16.5	−4.7	−10.0
<i>Realistic backtest — prediction (Rank IC)</i>										
MAIN-full	.155	.156	.132	.131	.055	.060	.033	.043	.061	—
ORIGINAL	.123	.126	.107	.110	.054	.053	.045	.048	.060	—
MAIN-kept	.127	.127	.109	.099	.047	.043	.019	.031	.056	—
<i>Benchmarks</i>										
<i>bench (EW, total)</i>	8.0	−26.6	35.8	27.3	7.5	−13.0	−4.9	21.3	20.0	6.6
<i>CSI300 (total)</i>	24.1	−23.8	38.6	29.8	−3.7	−20.4	−10.1	16.5	19.1	5.6

Table 6: Both backtests, 2017–2025, at **100M CNY** of capital. Returns are compound annual growth rates in %, so the “full” column is the rate that, compounded over the nine years, reproduces the actual result. The simple backtest does not correct for dividends, so its return rows compare between libraries rather than against the realistic ones. **Conclusion: main-full leads on both backtests and predicts better in all nine years; but the two disagree sharply after 2020 — profitable throughout under simple costs, negative from 2021 under realistic ones — and they disagree on prediction too, which points at the merging model rather than the formulas.**

n	10	20	40	56	80	120	full
MAIN-FULL Rank IC	0.0232	0.0344	0.0388	0.0452	0.0478	0.0483	0.0493
ORIGINAL Rank IC	0.0288	0.0384	0.0391	0.0441	0.0505	0.0501	0.0515

Table 7: Prediction quality against the number of formulas fed to the merging model, within a single library (2022–2025, 100M CNY, realistic cost model). **Conclusion: prediction rises with count alone, so libraries of different sizes cannot be compared directly — any comparison that does not hold the count fixed penalises the more selective system.**

Arm at $n = 56$	Rank IC	Return
MAIN gated zoo (the actual book)	0.0375	-5.28%
ORIGINAL subsets (mean $\pm$ sd)	0.0395 $\pm$ 0.0011	-7.43%
MAIN-full subsets (mean $\pm$ sd)	0.0462 $\pm$ 0.0021	-4.80%

Table 8: Like-for-like comparison at 56 formulas each (2022–2025, 100M CNY, realistic cost model), using five random draws from the two 150-formula libraries. **Conclusion: at equal size main’s generator wins clearly (+0.007 prediction, +2.6 points of return, no overlap across draws); its quality gate beats the baseline on return but roughly ties a random draw from main’s own library.**

System	Measure	Change per 100 formulas	First third $\rightarrow$ last third	p
MAIN	Rank IC	+0.0236	0.0311 $\rightarrow$ 0.0545 (+75%)	< 0.001
MAIN	IC	+0.0126	0.0080 $\rightarrow$ 0.0209 (+161%)	< 0.001
ORIGINAL	Rank IC	+0.0038	0.0324 $\rightarrow$ 0.0357 (+10%)	< 0.001
ORIGINAL	IC	+0.0030	0.0144 $\rightarrow$ 0.0165 (+15%)	< 0.001
MAIN	Rank IC of the kept set	+0.0617	0.0314 $\rightarrow$ 0.0545	< 0.001

Table 9: Improvement in formula quality across the 150 formulas each system mined, each scored on its own recorded numbers so only slopes are compared, never levels. Every trend survives a permutation test against shuffled ordering. **Conclusion: both systems learn, but main learns roughly six times faster (+0.0236 against +0.0038 per hundred formulas), and the formulas it keeps improve faster still.**

Operator	Metric	MAIN	ORIGINAL	chance	n (MAIN/ORIGINAL)
mutation	Rank IC	61.3% ($p=0.049$)	27.6%	50%	62 / 29
mutation	Rank ICIR	56.5%	27.6%	50%	62 / 29
mutation	IC	50.0%	31.0%	50%	62 / 29
crossover	IC	58.1%	21.1%	33%	43 / 19
crossover	Rank IC	55.8%	26.3%	33%	43 / 19
crossover	ICIR	55.8%	21.1%	33%	43 / 19

Table 10: How often a newly bred formula beats the one(s) it was bred from, on each system’s own recorded prediction scores. **Conclusion: main’s breeding consistently beats chance (53–61% against bars of 33–50%), while original’s falls below it (21–28%) — its breeding usually destroys value.**

	MAIN	ORIGINAL
(L1) does not get worse	PASS — rises, $p<0.001$	PASS — rises, $p<0.001$
(L2) the kept set improves	PASS — +0.062 per 100 kept	N/A — keeps everything
(L3) the best keeps rising	PASS — 0.023 $\rightarrow$ 0.058	PARTIAL — 0.031 $\rightarrow$ 0.042
Rate of improvement	+0.0236 per 100	+0.0038 per 100

Table 11: The three learning tests, measured per formula on each system’s own recorded prediction quality (Rank IC). **Conclusion: main passes all three; original passes the first, partially passes the third, and cannot be scored on the second because it keeps every formula it produces.**

	100M CNY			500M CNY			1B CNY		
	held	compnd.	grew	held	compnd.	grew	held	compnd.	grew
<i>2017–2025, excess return (% a year)</i>									
MAIN-full	+10.18	+7.84	3.6×	+5.57	+2.16	2.2×	+2.25	−1.20	1.6×
MAIN-kept	+4.14	+2.76	2.3×	−0.30	−2.10	1.5×	−3.51	−5.03	1.1×
ORIGINAL	+3.60	+2.40	2.3×	−1.10	−2.47	1.5×	−4.48	−5.41	1.1×
<i>2017–2025, net return (% a year)</i>									
MAIN-full	+17.18	+14.69		+12.29	+8.65		+8.75	+5.09	
MAIN-kept	+10.71	+9.24		+5.99	+4.08		+2.58	+0.96	
ORIGINAL	+10.14	+8.87		+5.15	+3.69		+1.56	+0.56	
<i>Trading cost (basis points), compounding</i>									
MAIN-full	4.93	5.81		6.68	8.04		8.00	9.41	

Table 12: Annual return at three fund sizes, realistic-cost backtest, 2017–2025. “held” charges trading as though the fund never changes size; “compnd.” lets the fund grow at its own return and charges each day against its actual size; “grew” is how many times over the fund multiplied. *Excess* return subtracts the market (which returned 6.60% a year); *net* is the return on the money itself. **Conclusion: main-full has much the best capacity at every size, but letting the fund compound costs 1–3.5 points a year and moves its practical ceiling from beyond 1B to between 500M and 1B.**

	MAIN-kept (56)	MAIN-full (147)	ORIGINAL (149)
<i>Diversity / redundancy</i>			
Mean pairwise $ \rho $	0.375	0.215	0.190
Share of pairs $\geq \rho_{\text{bar}}$	15.5%	4.4%	5.1%
Effective rank	12.6	38.1	36.0
Rank density (per factor)	22%	26%	24%
Distinct operators used	25	28	25
Effective operators (evenness)	16.4	18.3	13.6
<i>Regime behaviour</i>			
Mean $ IC $, crash days	0.0484	0.0540	0.0576
Mean $ IC $, rally days	0.0460	0.0451	0.0430
Crash-to-overall ratio	5.52	4.35	4.43
Share whose edge survives a crash	89%	88%	90%

Table 13: Variety and behaviour in bad markets (2022–2025, 100M CNY of capital, realistic cost model). Similarity is judged with the same threshold the quality gate itself uses. “Crash” and “rally” days are the worst and best 20% of days for the market as a whole. **Conclusion: the variety claim holds — main carries more independent ideas and spreads them more evenly across building blocks — but the crash-robustness claim does not: original is marginally better on both crash measures.**

	ORIGINAL	MAIN
What gets mutated	everything, in index order	sorted by why it failed
Crossover pairs	variety + raw score	the two that helped most
Failure diagnosis	none	five named failure modes
Success diagnosis	none	what each parent did well
Fresh ideas mid-run	none	injected when progress stalls
Quality gate	none (complexity check only)	multi-layer, portfolio-based
Feedback	previous idea + free text	full scorecard + memory of past failures

Table 14: Both systems mutate and cross over; what differs is what those operators select on, and whether anything is diagnosed. **Conclusion: original selects positionally and diagnoses nothing, so no measured outcome can change a later proposal — it has no channel through which learning could occur.**